3

Motion

Kinematics

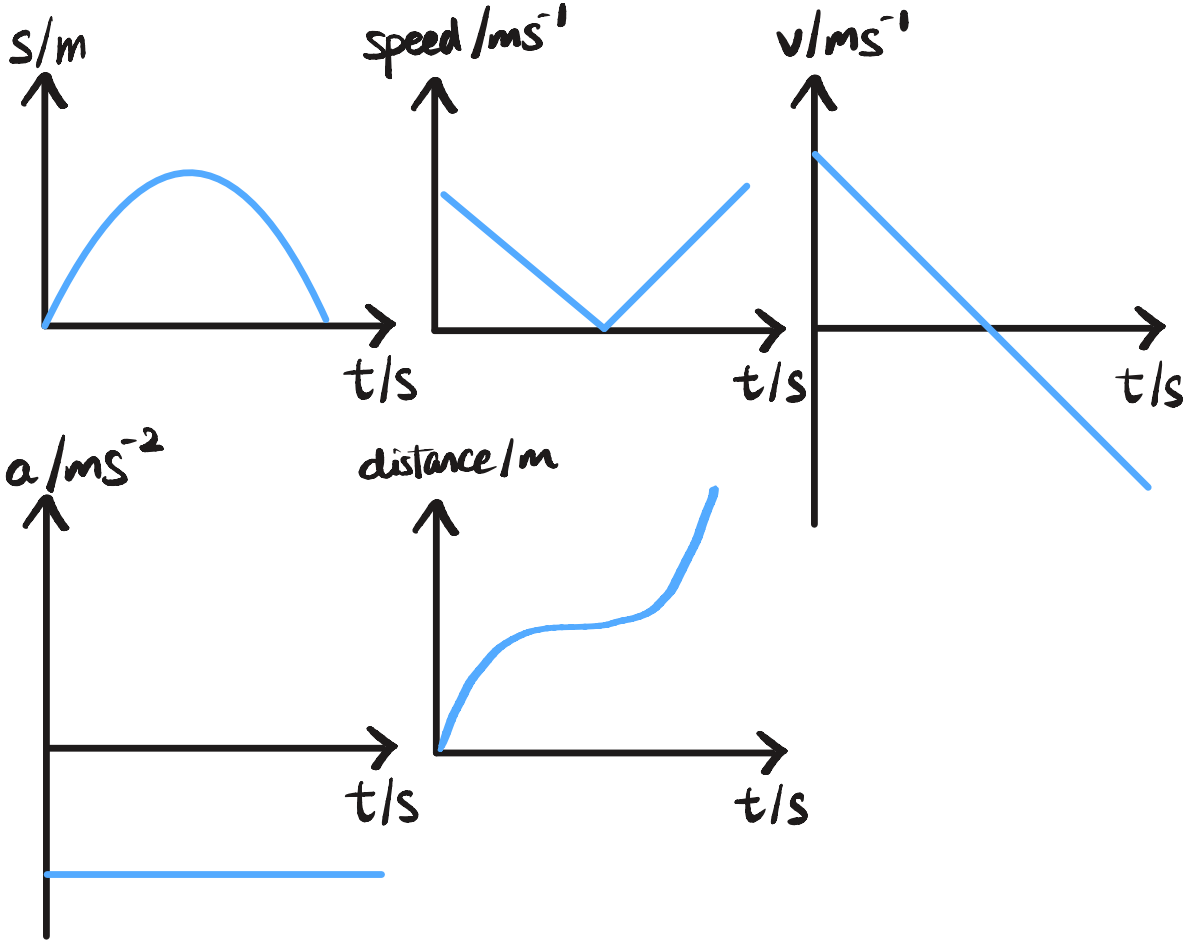
SUVAT

Projectile motion

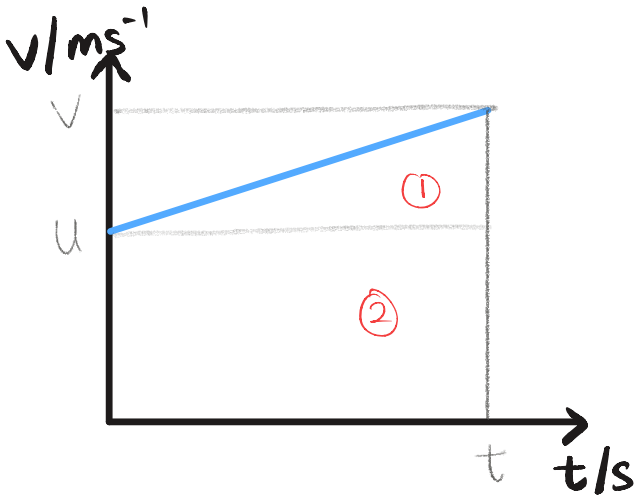
Kinematics

δ - small delta Δ - large delta

For a ball being thrown up, then falls back down
Air resistance is negligible



SUVAT



$$s = \int v\,dt$$

$$= ① + ②$$

$$= ut + \frac{1}{2}(v-u)t$$

$$= ut + \frac{1}{2}(at)t$$

$$s = ut + \frac{1}{2}at^2$$

$$u = v - at$$

$$s = (v-at)t + \frac{1}{2}at^2$$

$$= vt - at^2 + \frac{1}{2}at^2$$

$$s = vt - \frac{1}{2}at^2$$

$$a = \frac{dv}{dt} = \frac{v-u}{t}$$

$$\Rightarrow v - u = at$$

$$\Rightarrow = ut + \frac{1}{2}vt - \frac{1}{2}ut$$

$$= \frac{1}{2}ut + \frac{1}{2}vt$$

$$s = \left(\frac{u+v}{2}\right)t$$

$$s = ut + \frac{1}{2}(v-u)t$$

$$= u\left(\frac{v-u}{a}\right) + \frac{1}{2}(v-u)\left(\frac{v-u}{a}\right)$$

$$as = uv - u^2 + \frac{1}{2}(v-u)^2$$

$$= \cancel{uv} - u^2 + \frac{1}{2}v^2 - \cancel{uv} + \frac{1}{2}u^2$$

$$= \frac{1}{2}v^2 - \frac{1}{2}u^2$$

$$\Rightarrow v^2 - u^2 = 2as$$

Free fall - motion of an object where weight is the only fore acting on it
Exact value of g varies depending on altitude and location
it decreases noticeably at very high atltitude

- Assume initial velocity is zero
- Consider only motion under constant acceleration

Unless the questions say otherwise

Stopping distances:
Reaction times:
Lab: 0.1s
Applying brake: 0.67s

Thinking distance = speed x reaction time

Braking distance = $\frac{v^2 - u^2}{2a}$

Braking distance $\propto u^2$

A project is an object that has been subjected to an initial force that sets it moving
Travels subject to a single constant force
Usually the force is object's weight
This object is in free fall
Theoretically, projectile motion is independent of mass
(Assuming that air resistance is negligible)